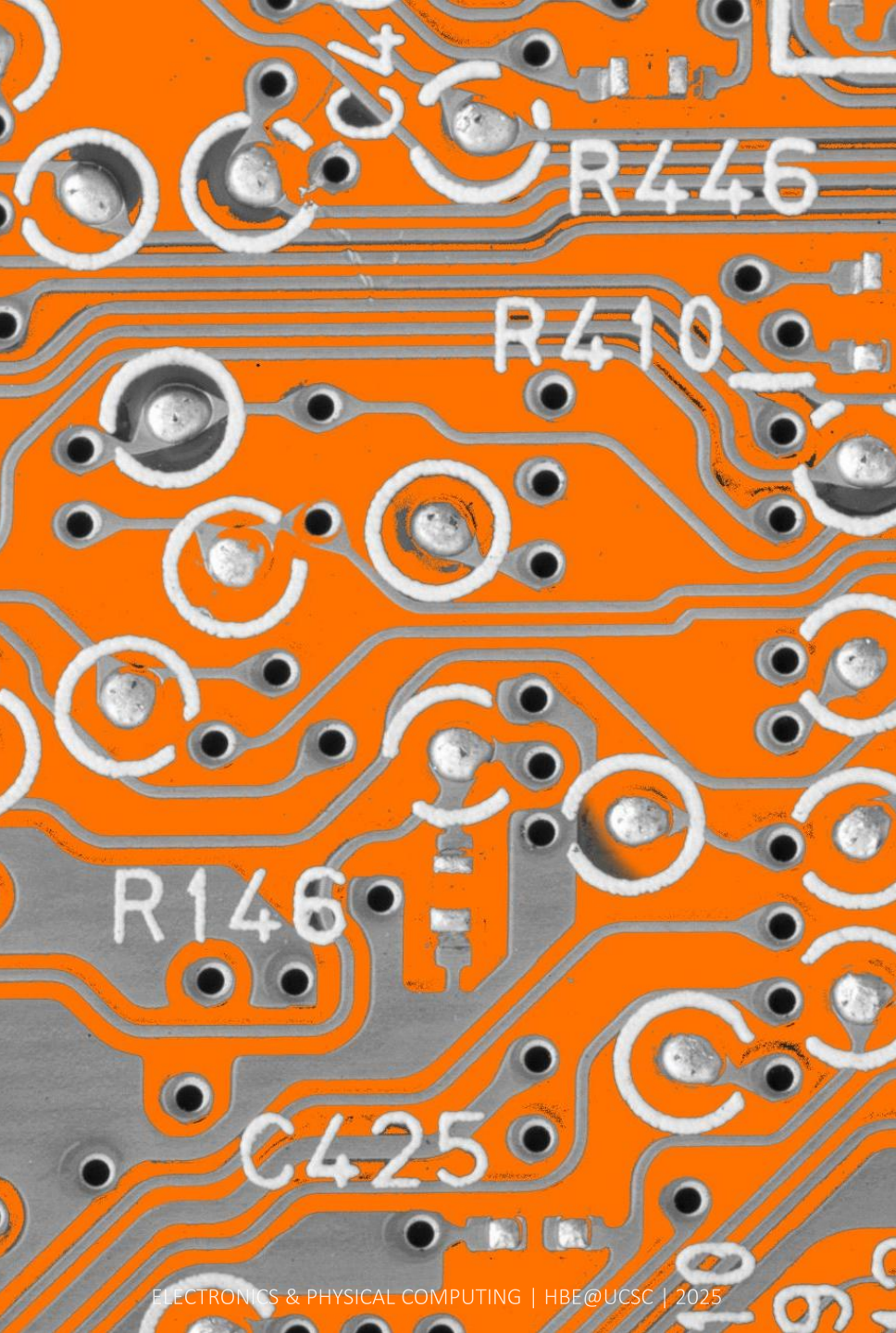


SCS2315 – Electronics & Physical Computing

Dr. H.E.M.H.B. Ekanayake

Lesson 1-2: Foundations of Electronics and Signals

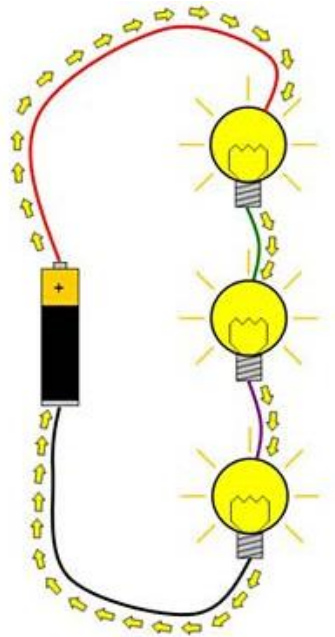
Electronics & Physical Computing



Circuit Types

Series Circuits

Components are connected **end-to-end**, so the **same current flows** through all of them.



🔑 Key Characteristics:

- **Current:** Same through all components
 $I_{\text{total}} = I_1 = I_2 = \dots$
- **Voltage:** Splits across components
 $V_{\text{total}} = V_1 + V_2 + \dots$
- **Resistance:** Adds up
 $R_{\text{total}} = R_1 + R_2 + \dots$

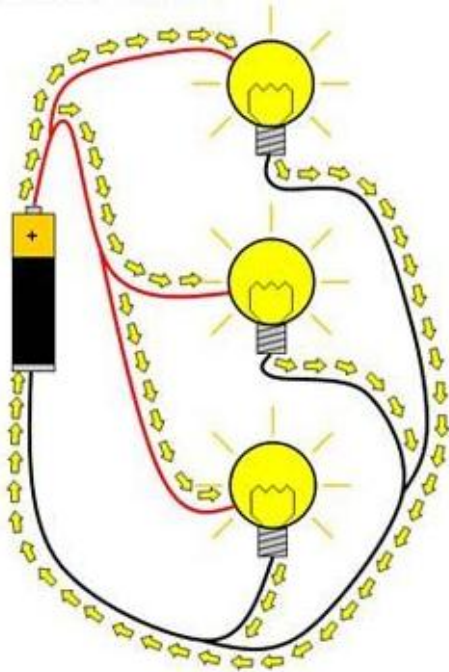
⚡ Example:

Two 2Ω resistors in series:

Total resistance = $2\Omega + 2\Omega = 4\Omega$

Parallel Circuits

Components are connected across the **same two nodes**, so the **same voltage** is applied to each.



🔑 Key Characteristics:

- **Voltage:** Same across all components

$$V_{\text{total}} = V_1 = V_2 = \dots$$

- **Current:** Splits between paths

$$I_{\text{total}} = I_1 + I_2 + \dots$$

- **Resistance:** Adds inversely

$$\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

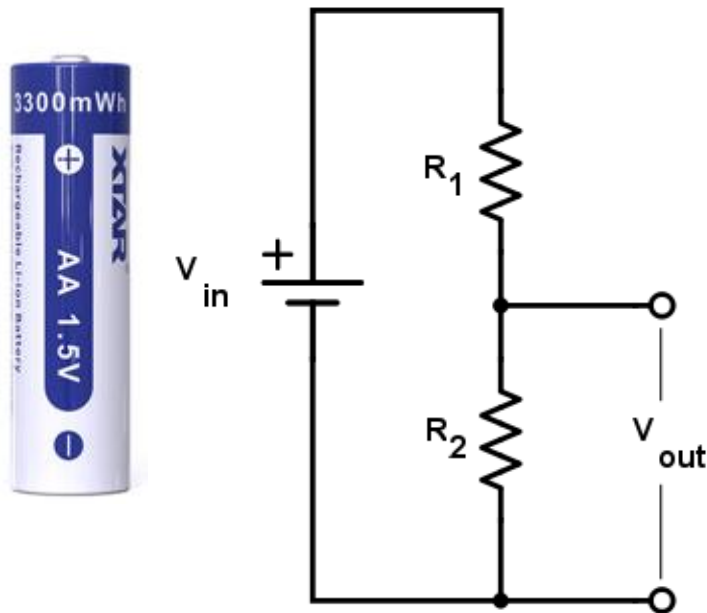
⚡ Example:

Two 2Ω resistors in parallel:

$$\frac{1}{R_{\text{total}}} = \frac{1}{2} + \frac{1}{2} = 1 \Rightarrow R_{\text{total}} = 1\Omega$$

Voltage Divider

A **series** circuit of resistors used to get a **reduced voltage** from a larger source.



 **Formula:**

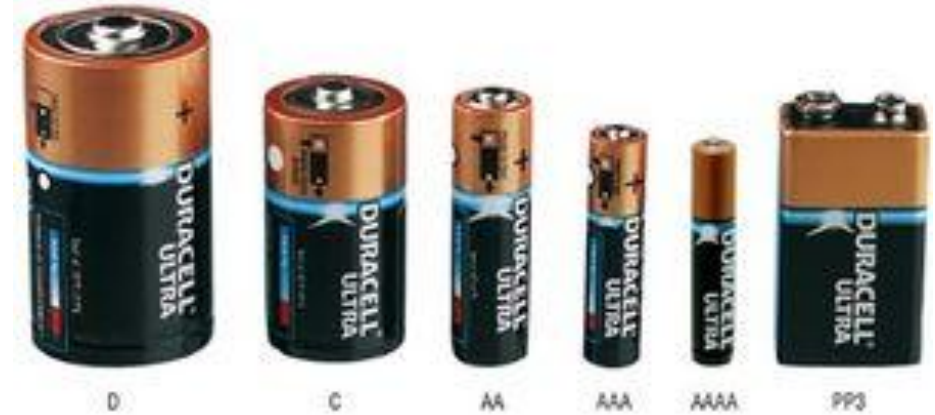
$$V_{out} = V_{in} \times \frac{R_2}{R_1 + R_2}$$

(Where V_{out} is across R_2)

 **Example:**

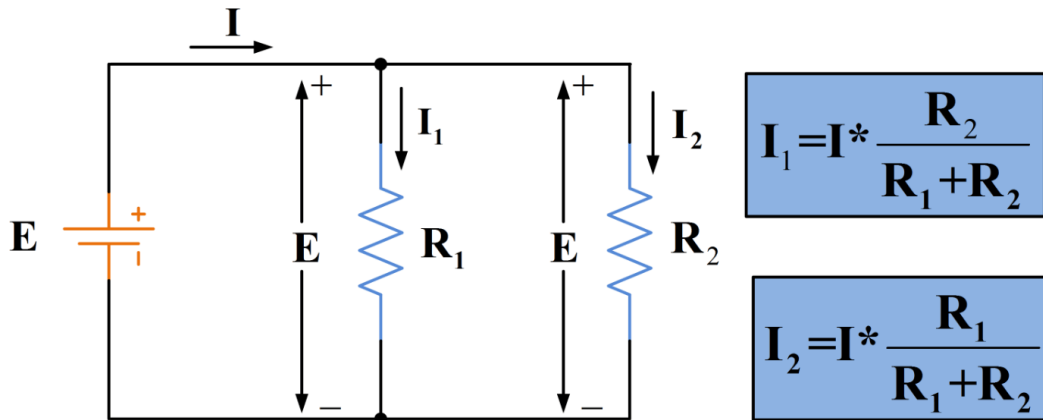
With 10V input, $R_1 = 2k\Omega$, $R_2 = 3k\Omega$:

$$V_{out} = 10 \times \frac{3000}{2000 + 3000} = 6V$$



Current Divider

A **parallel circuit** used to split current between branches.



 **Formula (for two resistors):**

$$I_1 = I_{\text{total}} \times \frac{R_2}{R_1 + R_2}$$

(Current through one resistor depends on the resistance of the *other*!)

 **Example:**

If $I_{\text{total}} = 4A$, $R_1 = 2\Omega$, $R_2 = 6\Omega$:

$$I_1 = 4 \times \frac{6}{2 + 6} = 3A$$

Exercise

Question:

A voltage divider consists of $R_1 = 3 \text{ k}\Omega$ and $R_2 = 6 \text{ k}\Omega$ across a 12 V source.

If a $6 \text{ k}\Omega$ load (R_L) is connected in parallel with R_2 , find the **loaded output voltage** across R_L .

Solution:

1. Combine R_2 and R_L (parallel):

$$R_{eq} = \frac{R_2 R_L}{R_2 + R_L} = \frac{6 \times 6}{6 + 6} = 3 \text{ k}\Omega$$

2. Voltage divider now has $R_1 = 3 \text{ k}\Omega$ and $R_{eq} = 3 \text{ k}\Omega$:

$$V_{out} = V_T \frac{R_{eq}}{R_1 + R_{eq}} = 12 \times \frac{3}{6} = 6 \text{ V}$$

✅ **Loaded output voltage = 6 V** (dropped from 8 V due to loading)

Summary

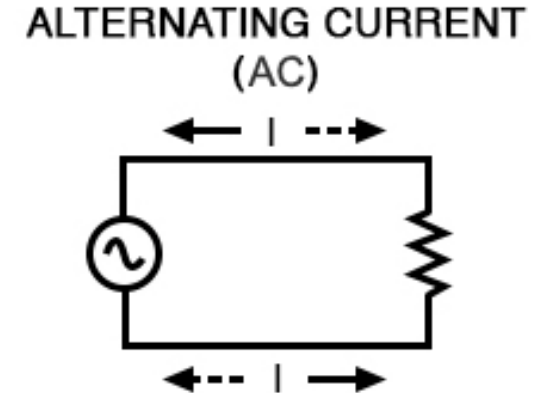
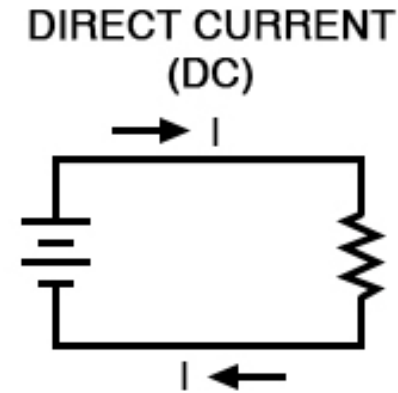
Type	Voltage	Current	Resistance
Series	Divides	Same	Adds up
Parallel	Same	Divides	Adds inversely
Voltage Divider	Drops voltage	—	Based on resistor ratio
Current Divider	—	Splits current	Based on inverse resistance



Signal Types

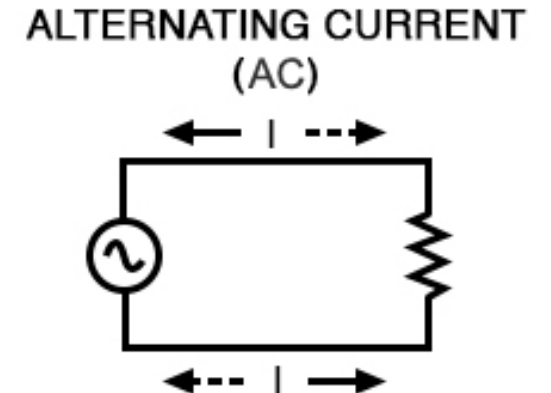
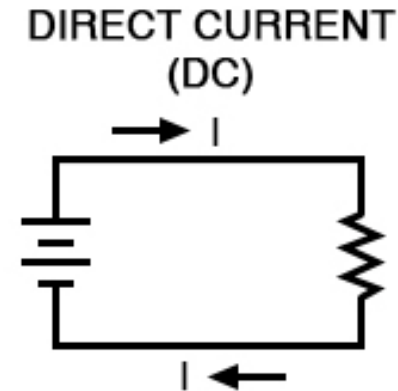
DC (Direct Current)

- **Definition:** Current flows in **one direction** only.
- **Source:** Batteries, solar cells, USB power.
- **Waveform:** Flat, constant line (like 5V or 3.3V steady).
- **Used in:** Electronics, microcontrollers (Arduino, Raspberry Pi), charging devices.



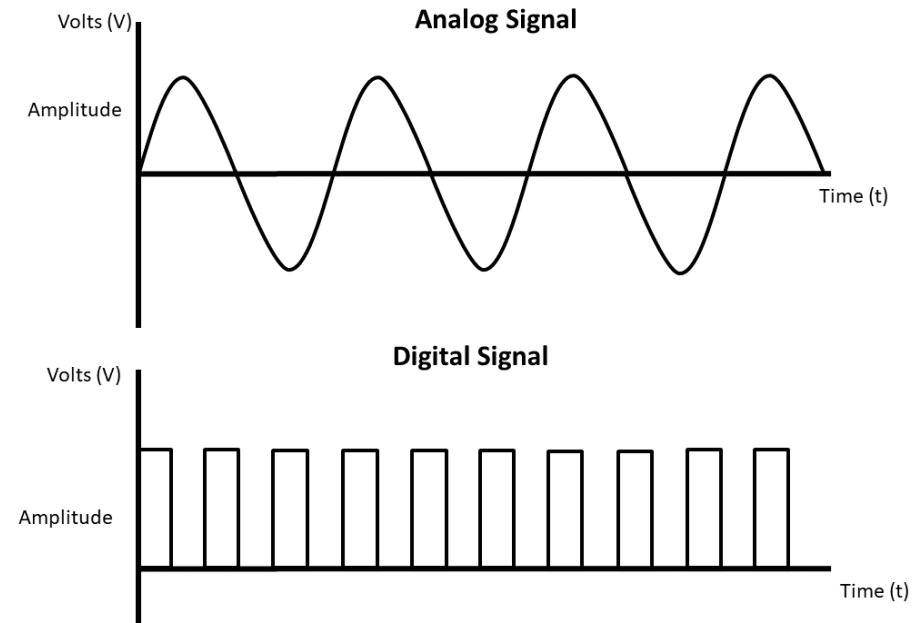
AC (Alternating Current)

- **Definition:** Current **reverses direction** periodically.
- **Source:** Power grid, wall outlets.
- **Waveform:** **Sine wave** (or sometimes square/triangular).
- **Frequency:** 50Hz (Europe, Asia), 60Hz (US).
- **Used in:** Household appliances, electric motors, audio signals.



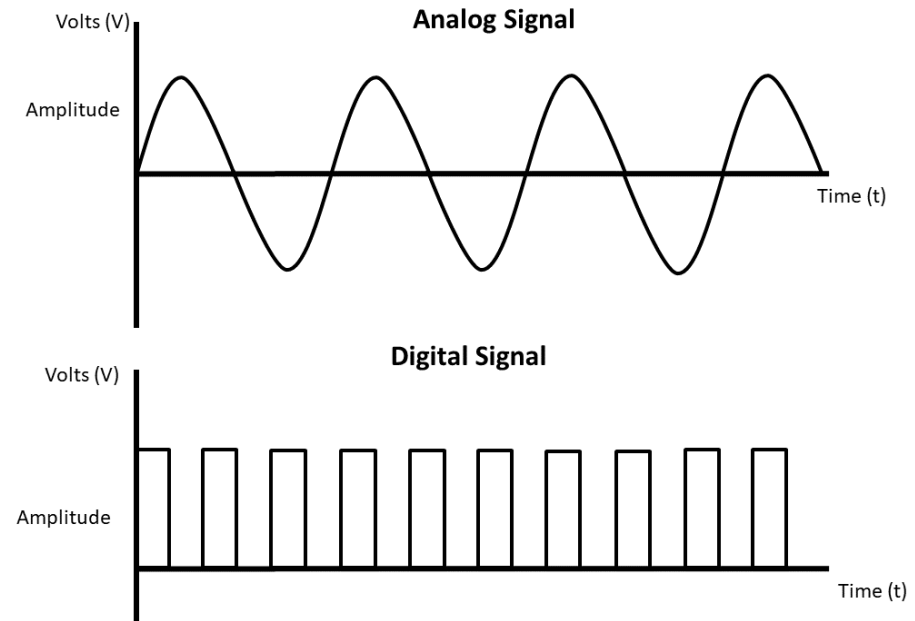
Analog Signal

- **Definition:** Varies **continuously** over time.
- **Range:** Infinite values within a range.
- **Example:** Sound wave, light intensity, temperature.
- **Waveform:** Smooth curves (like a sine wave).
- **Used in:** Sensors, microphones, analog audio/video.



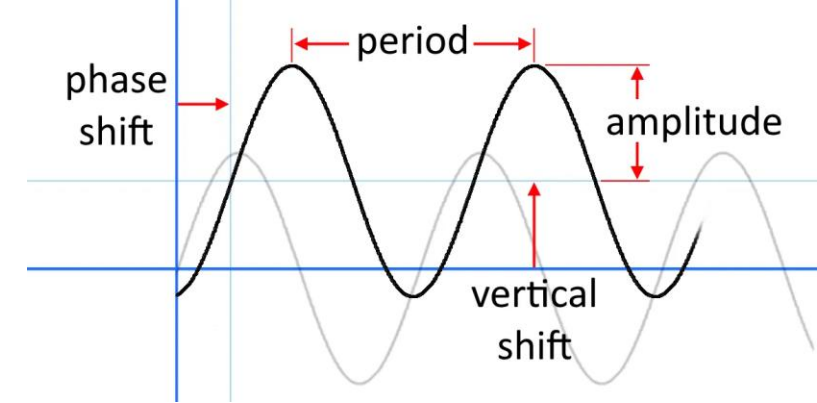
Digital Signal

- **Definition:** Varies in **discrete steps**, often just **0 and 1** (binary).
- **Range:** Finite number of values.
- **Example:** Computer data, on/off switches.
- **Waveform:** **Square wave** (sharp jumps between levels).
- **Used in:** Computers, microcontrollers, digital communication.



Signal Properties

Amplitude



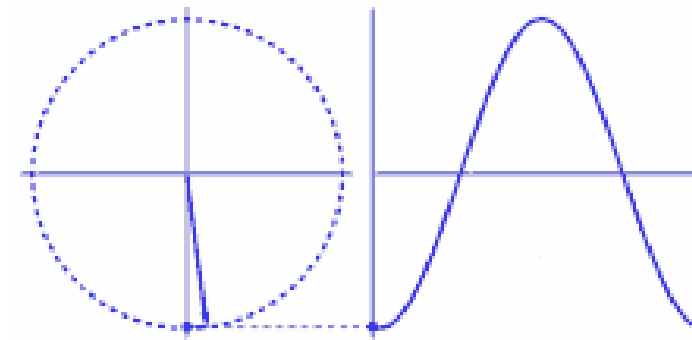
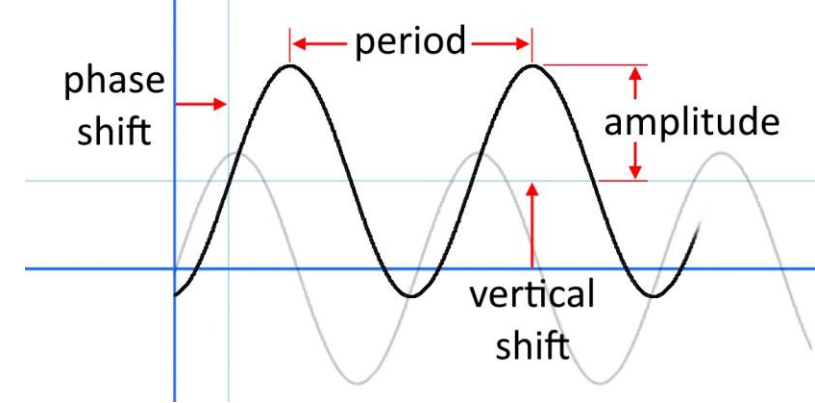
- **Definition:** The **maximum height** of the wave from its center (zero level).
 - **Unit:** Volts (V)
 - **Types:**
 - **Peak amplitude:** From 0 to the top.
 - **Peak-to-peak:** From bottom to top.
- 🔧 **Example:** A sine wave going from -5V to +5V has a **peak amplitude** of 5V and a **peak-to-peak** value of 10V.

Frequency (f)

- **Definition:** How often a wave repeats per second.
- **Unit:** Hertz (Hz) = cycles per second
- **Formula:**

$$f = \frac{1}{T}$$

🔧 **Example:** If a signal repeats 1000 times in a second, its frequency is **1 kHz**.



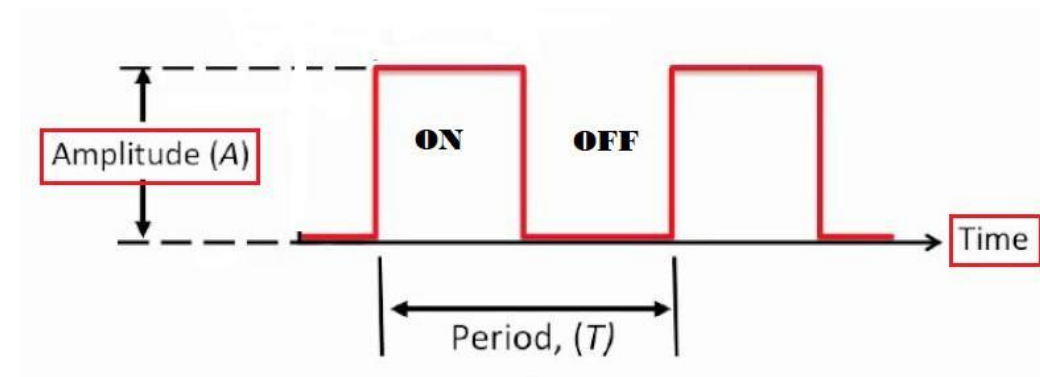
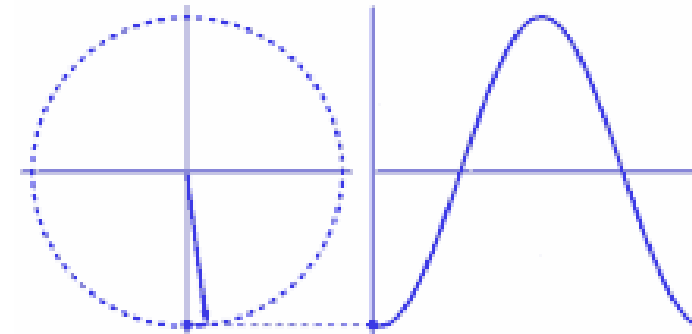
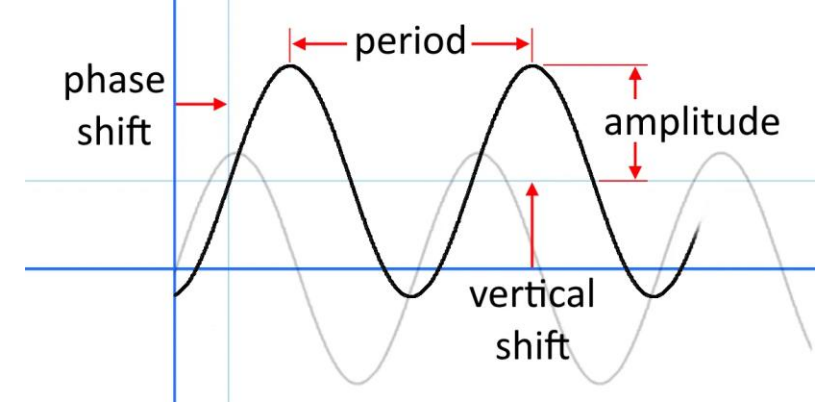
Period (T)

- **Definition:** The time it takes to complete one full cycle of the wave.
- **Unit:** Seconds (s)
- **Formula:**

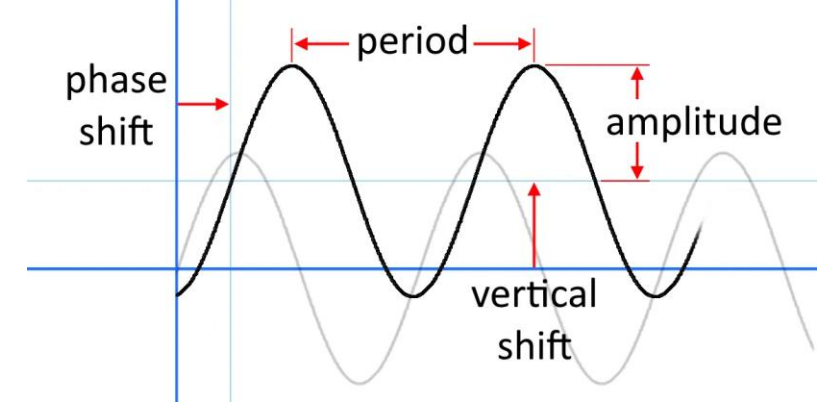
$$T = \frac{1}{f}$$

🔧 **Example:** A 50 Hz signal has a period of:

$$T = \frac{1}{50} = 0.02 \text{ s} = 20 \text{ ms}$$

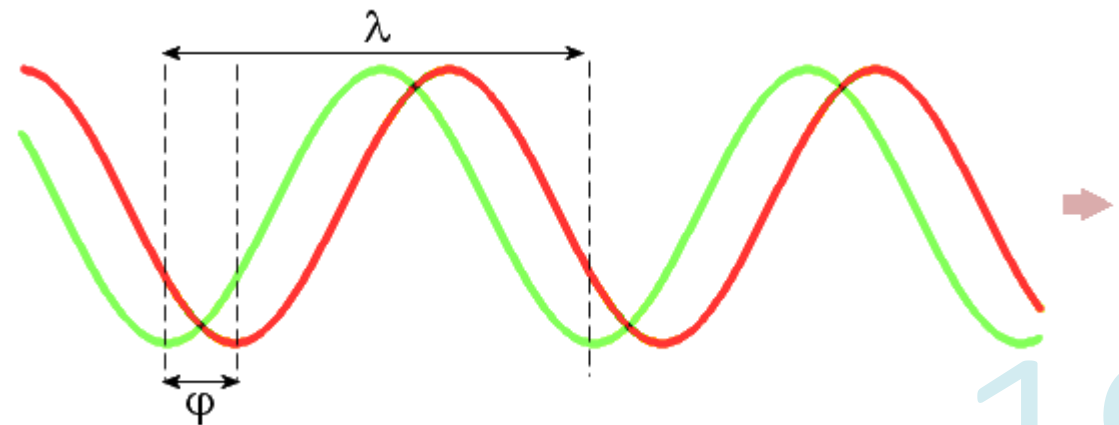
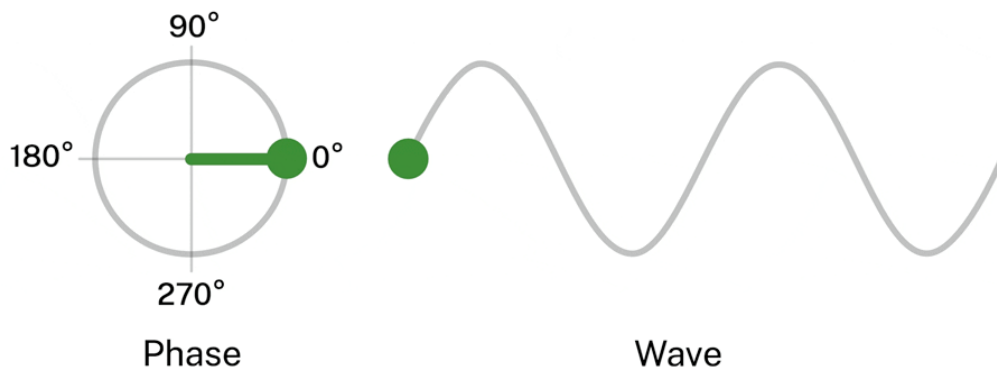


Phase (Φ)



- **Definition:** The **offset or shift** of a waveform in time, relative to another waveform.
- **Unit:** Degrees ($^{\circ}$) or radians
- **Common Uses:** Comparing waveforms (e.g., two sine waves out of phase by 90°)

🔧 **Example:** If two waves are identical but one starts a bit later, it's said to have a **phase shift** (e.g., 90°).



Sine Wave Equation

The general **sine wave equation** of an AC (alternating current) signal is:

$$v(t) = V_m \sin(\omega t + \phi)$$

- $v(t)$ → instantaneous voltage at time t
- V_m → maximum (peak) voltage
- ω → angular frequency in radians per second ($\omega = 2\pi f$)
- f → frequency in hertz (Hz)
- ϕ → phase angle in radians (initial phase shift)
- t → time in seconds

If an AC voltage has a peak value of 10 V, frequency 50 Hz, and no phase shift, then:

$$v(t) = 10 \sin(2\pi \times 50t)$$

Exercise #1

Question:

An AC voltage has a frequency of 60 Hz. Find its angular frequency ω .

Answer:

$$\omega = 2\pi f = 2\pi(60) = 120\pi \text{ rad/s} \approx 377 \text{ rad/s}$$

Question:

The AC voltage is given by $v(t) = 100 \sin(100\pi t)$.

Find the instantaneous voltage at $t = 3 \text{ ms}$.

Answer:

$$v(t) = 100 \sin(100\pi \times 0.003) = 100 \sin(0.3\pi)$$

$$v(t) = 100 \times 0.809 = 80.9 \text{ V}$$

Exercise #2

Question:

Given $v(t) = 25 \sin(200\pi t)$, determine:

1. The peak voltage
2. The frequency

Answer:

1. $V_m = 25 \text{ V}$
2. $\omega = 200\pi = 2\pi f \Rightarrow f = 100 \text{ Hz}$

Question:

The voltage is $v(t) = 50 \sin(314t + 45^\circ)$.

Find the:

1. Frequency
2. Instantaneous voltage at $t = 2 \text{ ms}$

Answer:

1. $\omega = 314 = 2\pi f \Rightarrow f = 50 \text{ Hz}$
2. Convert 45° to radians: $\phi = \pi/4$

$$v(t) = 50 \sin(314 \times 0.002 + \pi/4) = 50 \sin(0.628 + 0.785) = 50 \sin(1.413)$$

$$v(t) = 50 \times 0.987 = 49.3 \text{ V}$$

Exercise #3

Question:

Given $v(t) = 25 \sin(200\pi t)$, determine:

1. The peak voltage
2. The frequency

Answer:

1. $V_m = 25 \text{ V}$
2. $\omega = 200\pi = 2\pi f \Rightarrow f = 100 \text{ Hz}$

Question:

The voltage is $v(t) = 50 \sin(314t + 45^\circ)$.

Find the:

1. Frequency
2. Instantaneous voltage at $t = 2 \text{ ms}$

Answer:

1. $\omega = 314 = 2\pi f \Rightarrow f = 50 \text{ Hz}$
2. Convert 45° to radians: $\phi = \pi/4$

$$v(t) = 50 \sin(314 \times 0.002 + \pi/4) = 50 \sin(0.628 + 0.785) = 50 \sin(1.413)$$

$$v(t) = 50 \times 0.987 = 49.3 \text{ V}$$

Exercise #4

Question:

An AC voltage is given by $v(t) = 141.4 \sin(100\pi t)$.

Find the RMS (root mean square) voltage.

Answer:

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}} = \frac{141.4}{1.414} = 100 \text{ V}$$

Question:

Two signals are given as:

$v_1(t) = 50 \sin(314t)$ and $v_2(t) = 50 \sin(314t + 90^\circ)$.

Which signal leads and by how much?

Answer:

- $v_2(t)$ has a $+90^\circ$ phase shift, meaning it leads $v_1(t)$ by 90° .

Exercise #5

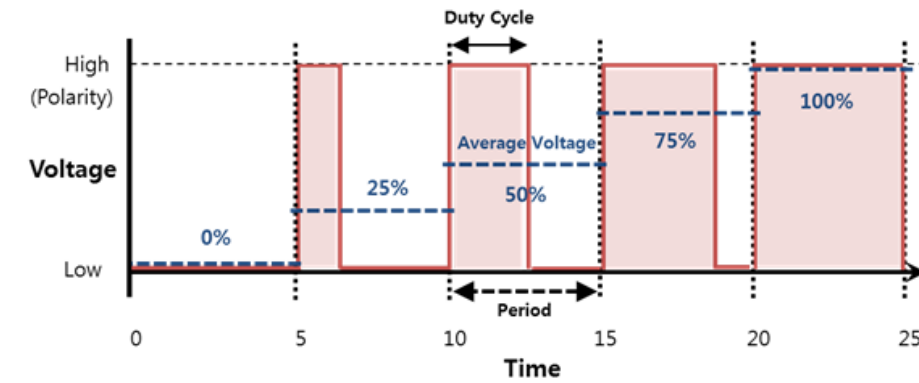
Question:

For $v(t) = 10 \sin(100\pi t)$, find the time t when $v(t)$ first becomes zero after $t = 0$.

Answer:

$$v(t) = 0 \text{ when } \sin(100\pi t) = 0 \Rightarrow 100\pi t = \pi \Rightarrow t = \frac{1}{100} = 0.01 \text{ s}$$

Duty Cycle



- **Definition:** For a **digital or square wave**, it's the **percentage of time the signal is HIGH (ON)** during one period.

- **Formula:**

$$\text{Duty Cycle} = \left(\frac{T_{\text{ON}}}{T_{\text{total}}} \right) \times 100\%$$

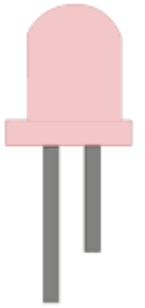
- **Unit: Percentage (%)**

🔧 Example:

- A 5V pulse wave is ON for 2 ms and OFF for 8 ms → Period = 10 ms
- Duty cycle = $\frac{2}{10} \times 100 = 20\%$



Duty Cycle: 0%





<https://www.youtube.com/watch?v=5nwNKPs2gco>

Duty Cycle Cont.

Signal Type	Description	Duty Cycle
Pure sine wave	Alternates symmetrically around zero	50%
Half-wave rectified sine	Positive half only	50% of full cycle (25% if full-wave considered)
Full-wave rectified sine	Two positive halves per cycle	100% (always positive)
PWM / square wave	ON/OFF waveform	Varies (0–100%)

Summary

Property	Describes	Unit	Example Value
Amplitude	Height of the wave	Volts (V)	5V peak
Frequency	How many cycles per second	Hertz (Hz)	60 Hz
Period	Time for one full cycle	Seconds (s)	0.02 s (20 ms)
Phase	Shift compared to another wave	Degrees (°)	90° out of phase
Duty Cycle	% of time signal is ON (in 1 cycle)	Percent (%)	50% (equal ON/OFF time)

RMS (Root Mean Square)

RMS is a way to express the **effective value** of a varying signal like an AC waveform. It tells us the **equivalent DC value** that would produce the **same heating effect (power)** in a resistor.

For AC signals (like sine waves), the voltage and current are always changing.

The **average voltage** of a full sine wave is actually zero (because the + and - cancel out).

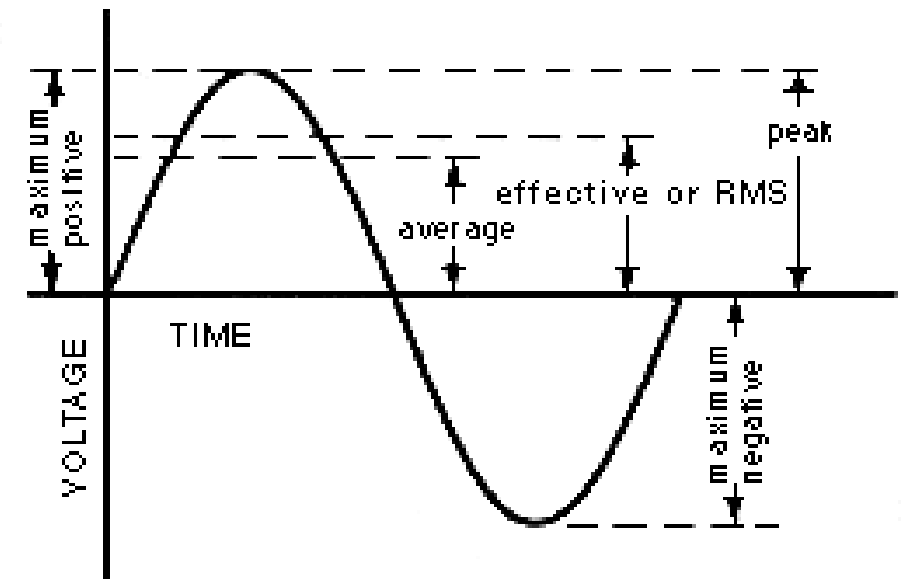
But that's not helpful when calculating things like power!

So instead, we use **RMS**, which gives us a kind of "average power value" that we can work with.

When you hear “230V AC” or “120V AC” — that's **RMS**, not peak.

RMS (Root Mean Square)

$$V_{\text{RMS}} = \frac{V_{\text{peak}}}{\sqrt{2}} \approx 0.707 \times V_{\text{peak}}$$



🔧 **Example:**

If an AC voltage has a peak of **100V**, the RMS voltage is:

$$I_{\text{RMS}} = \frac{I_{\text{peak}}}{\sqrt{2}}$$

$$V_{\text{RMS}} = \frac{100}{\sqrt{2}} \approx 70.7V$$

$$\text{Power} = V_{\text{RMS}} \times I_{\text{RMS}}$$

This means a **70.7V DC source** would deliver the **same power** to a resistor as the 100V AC source.



Time & Frequency Domain

Time Domain

- **Definition:** A view of a signal as it changes over time.
- This is the **default way** we usually see waveforms — like oscilloscopes show.
- You plot **amplitude (voltage or current)** vs. **time**.

Example:

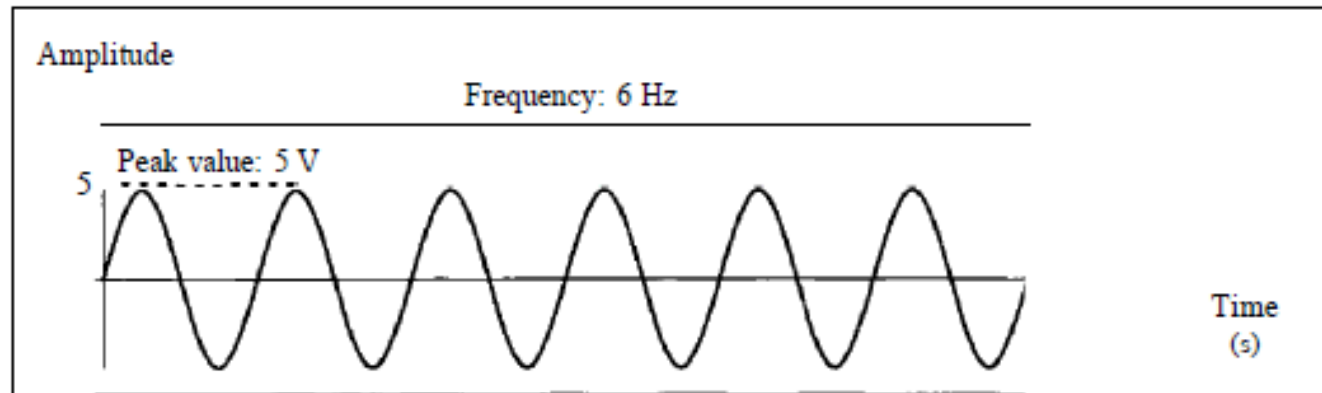
A sine wave that goes up and down smoothly over time — like a 50Hz AC signal.

Frequency Domain

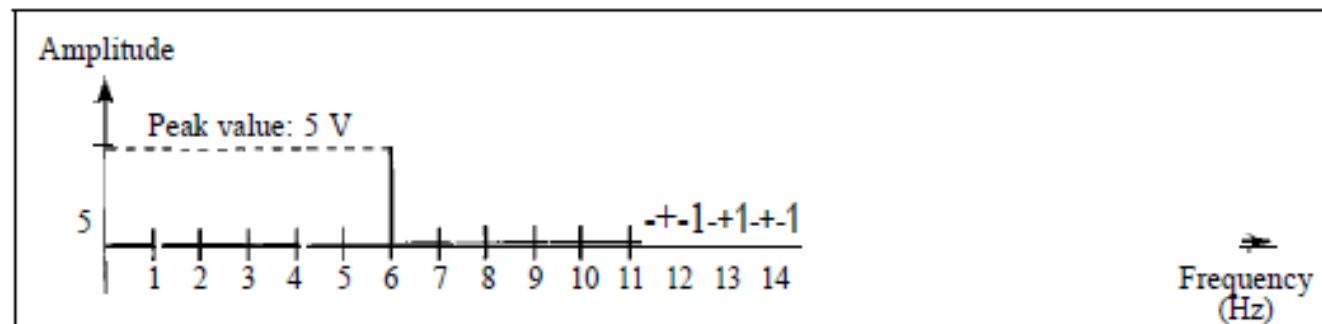
- **Definition:** Shows **which frequencies** are present in a signal and **how strong** they are.
- Instead of showing amplitude over time, we show **amplitude vs. frequency**.
- Useful for analyzing complex signals made of multiple frequencies (like audio, RF, or digital signals).

Example:

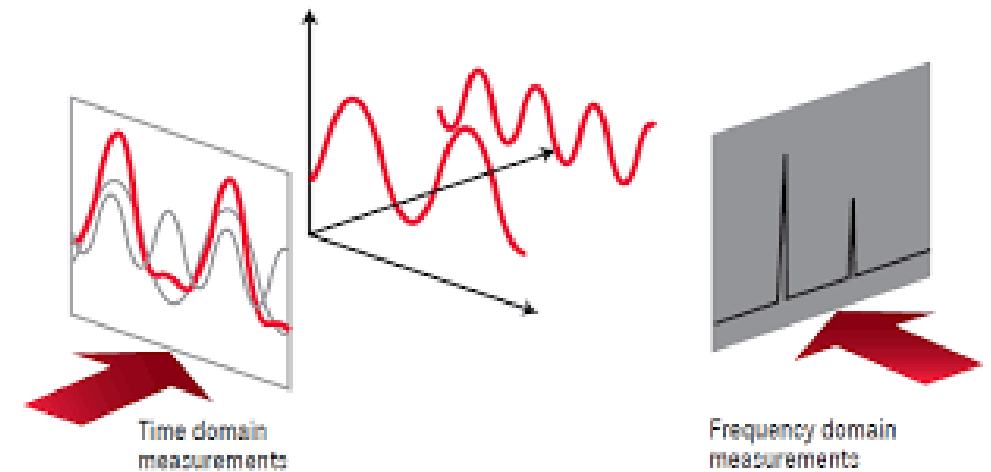
An audio signal might contain 440Hz (musical note A), 880Hz, and 1320Hz — the **frequency domain** tells you how much of each is present.



a. A sine wave in the time domain (peak value: 5 V, frequency: 6 Hz)

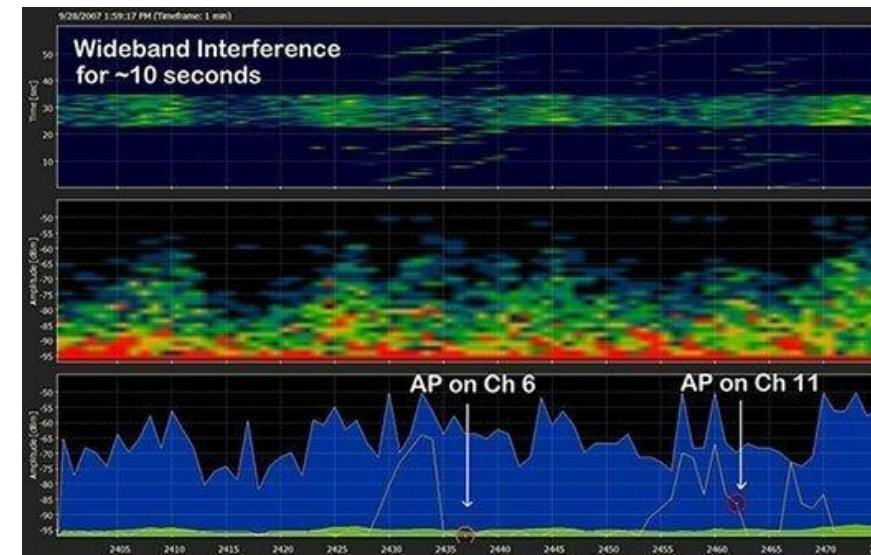


b. The same sine wave in the frequency domain (peak value: 5 V, frequency: 6 Hz)



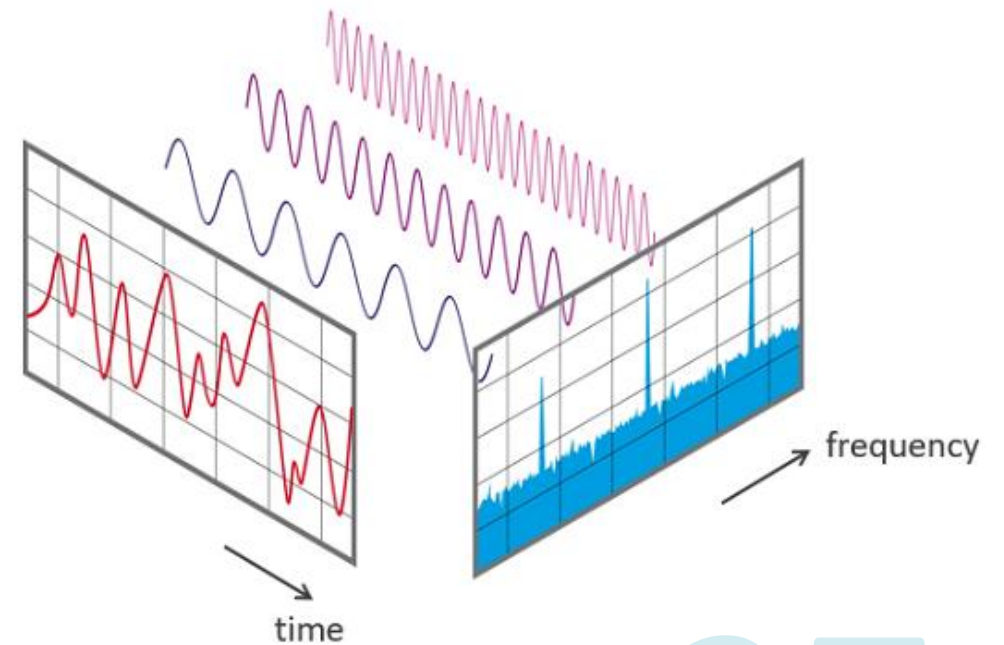
Spectrum Analysis

- The process of **analyzing** a signal's frequency content.
- The result is called the **frequency spectrum**.
- Think of it like “decomposing” a signal into its **building blocks** (frequencies).
- Tools like **spectrum analyzers** or **software FFTs** help visualize this.



FFT (Fast Fourier Transform)

- FFT is an **algorithm** that converts a signal from the **time domain** to the **frequency domain**.
- It's a **fast** and efficient version of the **Fourier Transform**.
- FFT shows:
 - What frequencies are in the signal
 - How strong each frequency is (amplitude)
 - Their **phase** (optional)



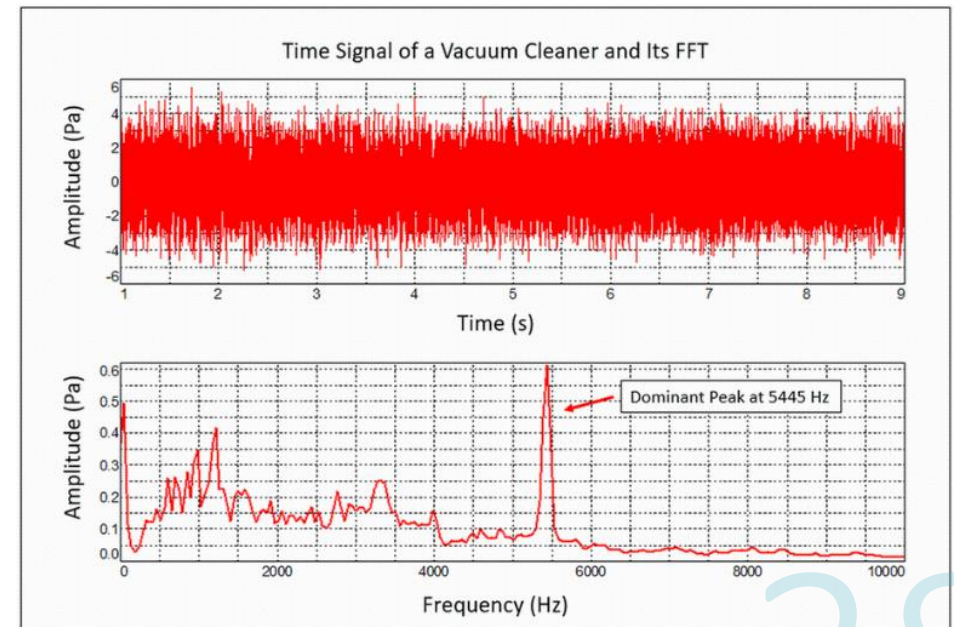
FFT (Fast Fourier Transform)



Example:

You record a sound wave (time domain) and apply an FFT — it tells you:

- There's a 1kHz tone
- A 5kHz spike
- Maybe some 50Hz noise



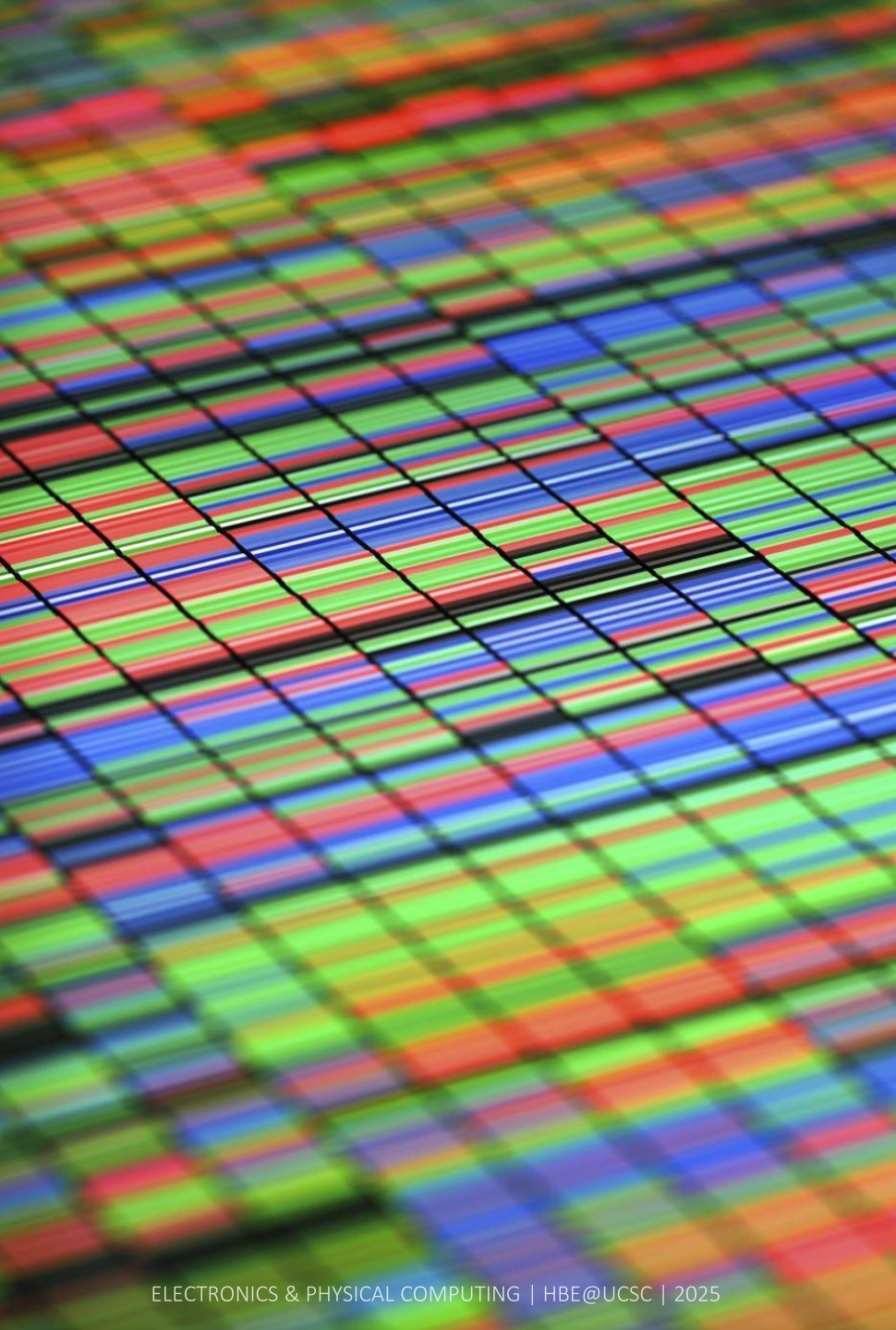
Comparison

Property	Time Domain	Frequency Domain
Shows	Amplitude over time	Amplitude vs frequency
Useful for	Timing, pulse width, wave shape	Filtering, modulation, noise analysis
Example Tool	Oscilloscope	Spectrum analyzer, FFT
Related Concept	Sampling, waveform	Harmonics, spectrum, bandwidth



Real-Life Analogy: Music

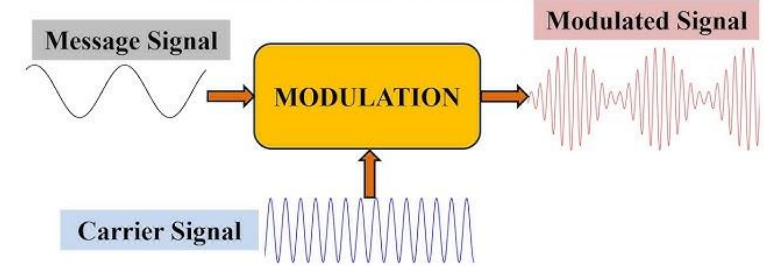
- **Time Domain:** You hear a song as a waveform (sound pressure changing over time).
- **Frequency Domain:** The same song as a mix of frequencies — bass, vocals, treble — with different intensities.



Signal Modulation

Modulation

What is Modulation?



Signal **modulation** is a technique used to transmit data over a medium (like airwaves or cables) by changing certain properties of a **carrier wave**.

It's the process of **altering a carrier signal** (typically a sine wave) to **encode information**.

Carrier wave: usually high-frequency and constant.

The information (data) modifies the **amplitude**, **frequency**, or **phase** of this wave.

ASK – Amplitude Shift Keying

- **How it works:** The **amplitude (height)** of the carrier wave changes based on data.

- Typically:

- Logic 1 → High amplitude
- Logic 0 → Low or zero amplitude



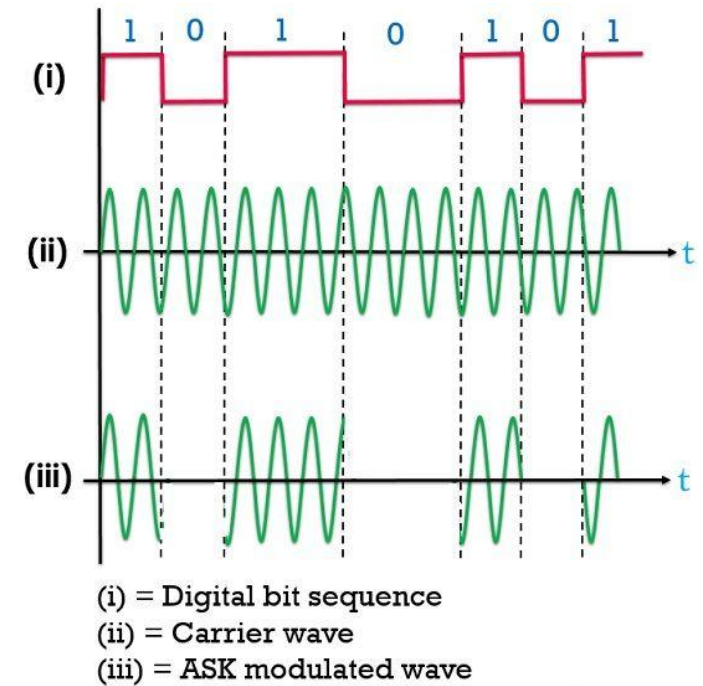
Characteristics:

- Simple to implement
- Sensitive to noise (especially amplitude noise)



Use Case:

- Low-speed modems
- Optical fiber systems (on-off light = ASK)



Electronics Coach

FSK – Frequency Shift Keying

- **How it works:** The **frequency** of the carrier changes with the digital data.
- Typically:
 - Logic 1 → Higher frequency
 - Logic 0 → Lower frequency



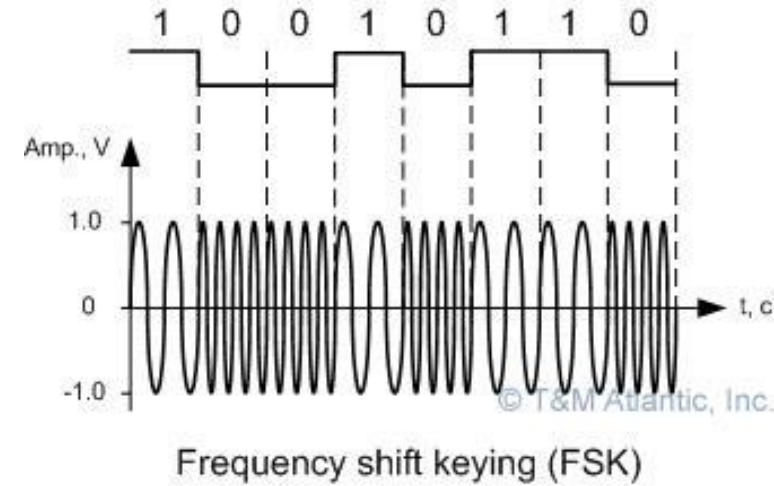
Characteristics:

- More noise-resistant than ASK
- Widely used in wireless communication



Use Case:

- Radio transmission
- Bluetooth, RFID, and older modems



PSK – Phase Shift Keying

- **How it works:** The **phase** of the carrier wave is shifted to represent data.

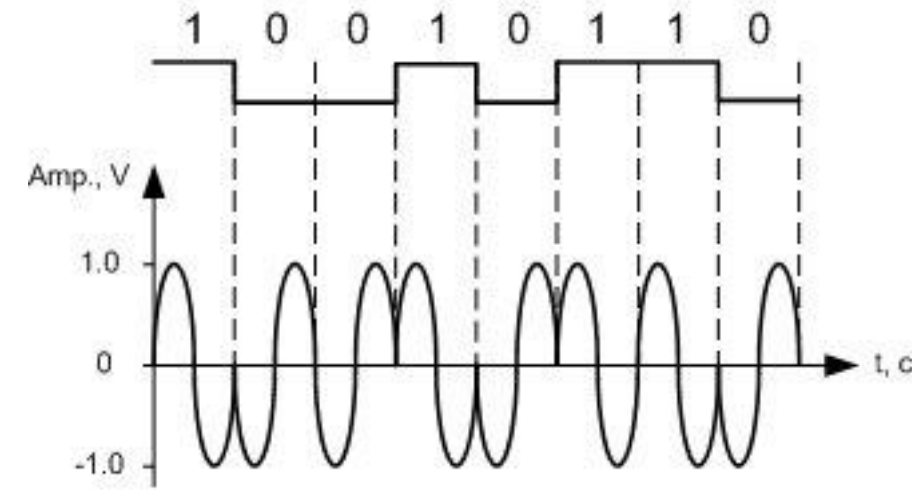
- For example:
 - Logic 1 → 180° phase shift
 - Logic 0 → No phase shift

Variants:

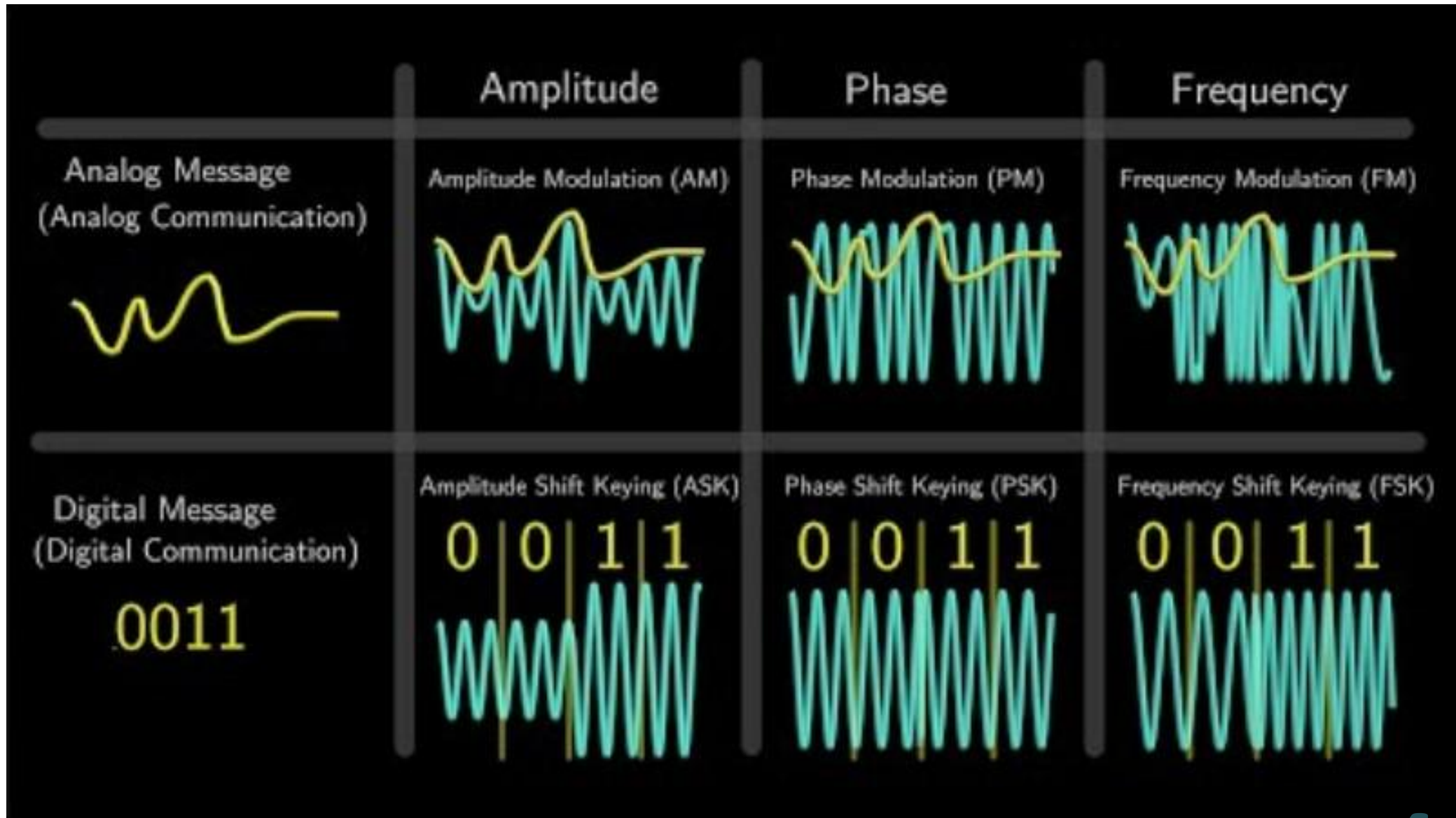
- **BPSK:** Binary Phase Shift Keying (2 phases: 0°, 180°)
- **QPSK:** Quadrature PSK (4 phases: 0°, 90°, 180°, 270° → carries 2 bits per symbol)

Use Case:

- Wi-Fi, satellite communication, 4G/5G, GPS



Phase shift keying (PSK)



<https://www.youtube.com/watch?v=c3eMoHuPRy0>

Modulation	Main Idea	Bits per Symbol	Typical Use
ASK	Turn the amplitude up/down	1	Simple systems, low speed
PSK	Flip or rotate the phase	1–3 (BPSK, QPSK, etc.)	Wireless, Wi-Fi
FSK	Switch between two or more frequencies	1–2	Modems, RF links
QAM	Combine amplitude + phase changes	4–8+	High-speed data (e.g., 16-QAM = 4 bits/symbol)

Comparison

Noise and Signal Integrity

Source	Description
Thermal Noise	From random electron motion in resistors/components
EMI (Electromagnetic Interference)	From nearby electronics, motors, power lines
Crosstalk	Signal leaking from adjacent wires/tracks
Power Supply Noise	Ripple or fluctuation in DC power
Switching Noise	From digital devices turning on/off rapidly
Environmental	Lightning, RF transmissions, fluorescent lights

What is Noise?

Noise is any unwanted electrical **signal** that interferes with the main signal. It can distort, degrade, or corrupt your data or measurements.

What is Signal Integrity?

Signal Integrity (SI) refers to the **quality and reliability** of the signal as it travels through a circuit or system.

If signal integrity is poor:

- You get **data errors**

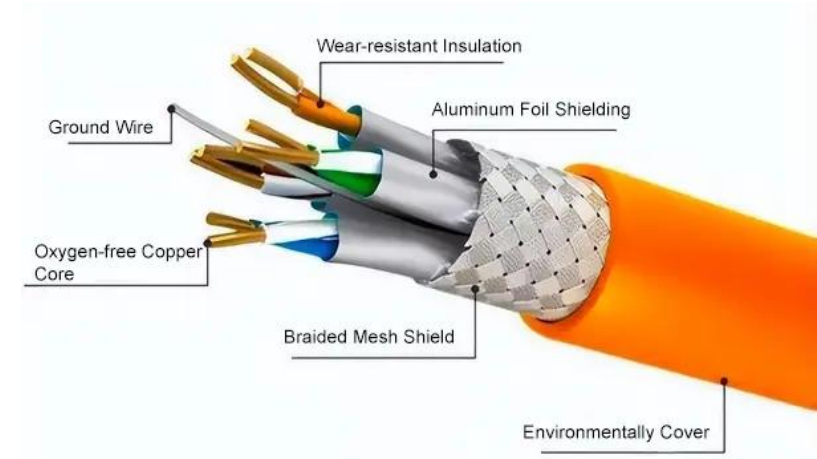
- The waveform may be **distorted**

- You may experience **timing issues** in digital systems

Techniques to Improve Signal Integrity

1 Shielding

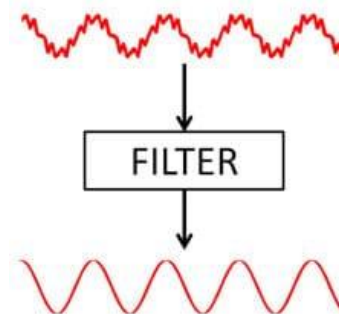
- **Purpose:** Block external electromagnetic noise from entering the signal path.
- **How:**
 - Use **shielded cables** (e.g., coaxial, twisted pair)
 - Add **metal enclosures or ground planes**
 - Route **signal and ground wires** together to reduce interference

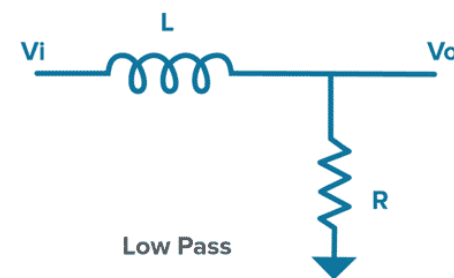
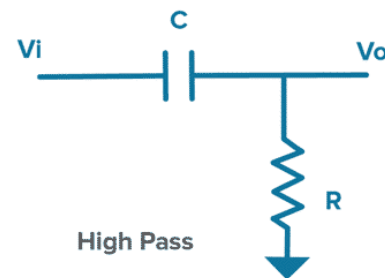
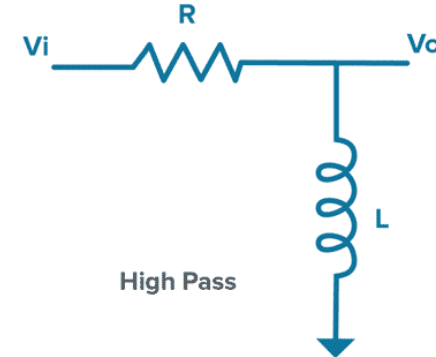
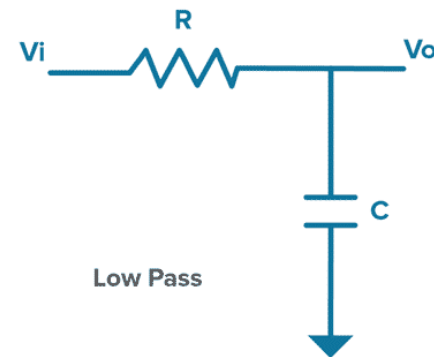
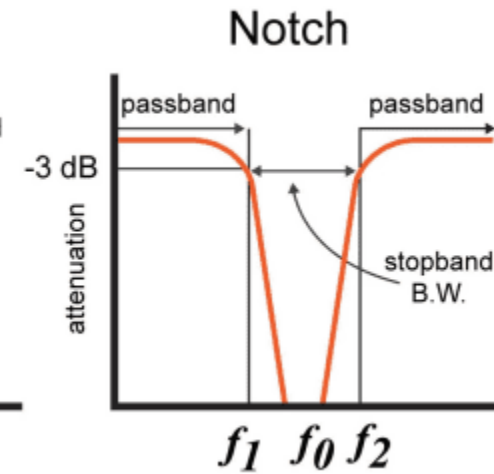
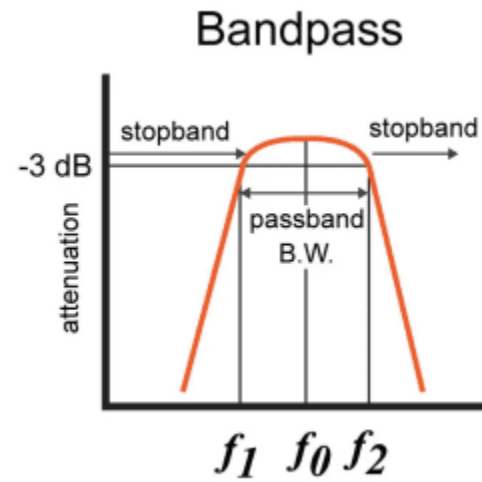
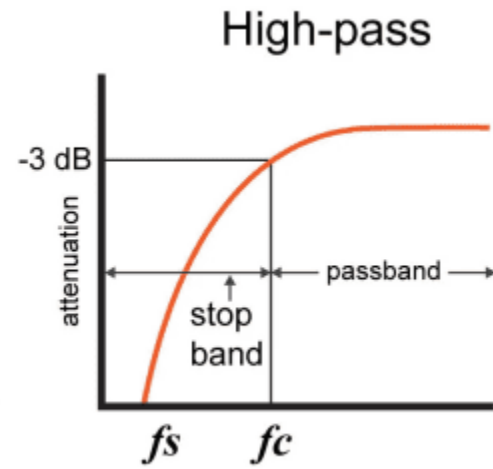
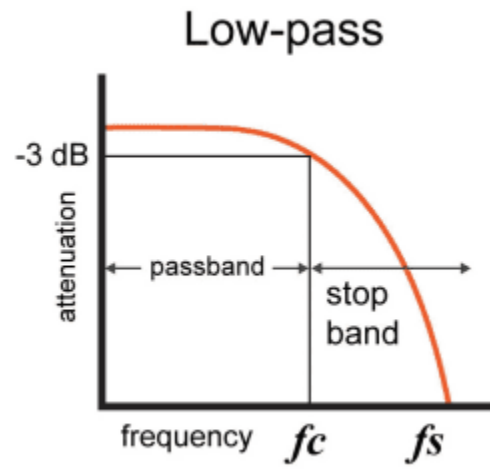


Techniques to Improve Signal Integrity

2 Filtering

- **Purpose:** Remove noise or unwanted frequency components.
- **Types:**
 - **Low-pass filters:** Allow DC or slow signals, block high-frequency noise
 - **High-pass filters:** Remove DC or low-frequency drift
 - **Band-pass filters:** Let only a range of frequencies pass
- **Components used:**
 - Resistors (R)
 - Capacitors (C)
 - Inductors (L)
 - Ferrite beads





Techniques to Improve Signal Integrity

3 Other Signal Integrity Practices

- **Proper grounding:** Avoid ground loops and ensure solid connections
- **Short signal paths:** Reduce antenna-like effects and EMI pickup
- **Use differential signals:** Like in USB, Ethernet, RS-485 — immune to common noise
- **Controlled impedance:** In high-speed PCBs, trace geometry is designed to prevent reflections
- **Decoupling capacitors:** Filter out power supply noise near chips

END of Lesson 1

Announcement

In-class assignment 1 will be held in the 2nd or 3rd week of December